

Kabir Singh

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ACADEMIC EDUCATION

- Albert Dorman Honors College (ADHC) & Jordan Hu College of Sciences & Liberal Arts** (JHCSLA), New Jersey Institute of Technology, Newark, New Jersey, USA Sep '23 – May '27 (Expected)
- Bachelor of Science; Major – **Mathematical Biology**
 - Minors – **Biomedical Engineering in Technology, Philosophy, History**
 - ADHC Honors Tracks – **Medical Humanities, Global Studies**
 - *Academic Excellence, Honors Merit, Honors Residential Scholarship Recipient (Tuition + Board)*
- Faculty of Engineering, Lunds Tekniska Högskola, **Lunds Universitet**, Lund, Sweden Jan '25 – Jun '25
- Fellow, Institute of International Education's Global Engineering Education Exchange (Global E3)

RESEARCH

- Undergraduate Researcher, Mathematical Neuroscience. PI: Dr. Horacio Rotstein, NJIT Sep '26 – Present
- Deriving reduced mathematical descriptions of biophysical neuronal models to characterize their dynamic properties
- Undergraduate Researcher, **History of Medicine & Bioethics**. PI: Dr. Stephen Pemberton, NJIT Sep '26 – Present
- Synthesizing legal, clinical, and cultural developments in American end-of-life care from the late twentieth century to the present
 - Developing original scholarship for submission to a peer-reviewed humanities journal
- Emerging Scholars Research Fellow**, Early Modern Sikh History. Mentor: Satnam Singh, Harvard Sikh Center Sep '26 – Present
- *Dissemination and the making of Khalsa authority in a contested panth*: examining how the Khalsa asserted dominance over the wider Sikh world within decades of its founding, against rival claimants including the Bandai Khalsa, Minas, and Ram Raiyas
 - Arguing that the consolidation of “authentic Sikh teachings” through katha was instrumental to that dominance; working from Gurmukhi and Persian primary sources, including works attributed to Bhai Mani Singh, hukamname, and rahit-nama literature
- Undergraduate Researcher, Computational Bionanotechnology. PI: Dr. Stefano Angioletti-Uberti, Imperial College London Jun '26 – Present
- **BAGEL: Biomolecular Algorithm for Guidance in Energy Landscape**
 - Computational modelling of soft-matter and biomaterial systems
 - Applying statistical mechanics, molecular dynamics, and machine-learning methods to problems in enzyme design and broader protein design
- Undergraduate Researcher, **Computational Biology Team Lead** at KumarLab. PI: Dr. Vivek Kumar, University of Houston (Prev. NJIT) Oct '23 – Present
- Lead undergraduate and graduate students on designing and developing peptide-based therapeutics
 - Started at NJIT and continued computational work with lab relocation to Houston
 - Taught 15+ associates in lab
 - Received > \$15,000 in direct funding for projects
- Undergraduate Researcher, **Mathematical Modeling**. PI: Dr. James Maclaurin, NJIT Oct '25 – May '26
- Applying Hawkes self-exciting point process to model secondary nucleation dynamics in amyloid formation
- Undergraduate Researcher, **Digital History & Urban Studies**. PI: Dr. Louis Hamilton, NJIT Jan '24 – May '24, Jan '26 – May '26
- Digital mapping of how religious institutions shaped urban form in medieval and early modern Italy; field research conducted in Rome and Florence
 - *Power vs. the People* (HIST 328; 18 sites): layered Counter-Reformation Marian shrines against Julius II's 1508 papal axis on the Via Giulia
 - *Processions and Power* (HIST 329; 22 sites): traced how Florence's expansion turned liturgical processions into contested displays of ecclesiastical and household prestige

FUNDING

- Undergraduate Research Opportunities Programme (UROP), Imperial College London Jun '26 – Jul '26
- Competitive research bursary awarded through the Department of Materials, Faculty of Engineering
 - Project: Using protein language models to mimic natural ligands. Supervisor: Prof. Stefano Angioletti-Uberti
 - Personal compensation included
- NSF I-Corps Regional Program, Northeast Hub May '26 – Jun '26
- Co-Founder & Technical Lead, Axioma Therapeutics
 - Selected into competitive NSF cohort; led customer discovery and lean-startup validation for cyclic peptide therapeutics targeting degenerative eye diseases
 - Project funding awarded to support customer-discovery activities
- Grace Hopper AI Research Institute, Computational Protein Design May '25 – Aug '25
- NJIT grant to develop a course on computational protein design
 - Teaching Assistant
- The Honors College Moonshot Grants for Engaged Scholarship Jun '25 – Jul '25
- Endowed award to develop interdisciplinary projects that address important social and cultural questions
 - Designing Male Contraceptive
 - Award providing labs with funds to support undergraduate-led projects
 - Personal compensation included. One recipient per annum
- Male Contraceptive Initiative (MCI) Jan '25 – Jun '25
- Award by the Youth Advisory Board providing funding to support grantee-led project
 - Designing Male Contraceptive
 - Personal compensation included
- NJIT Undergraduate Research & Innovation (URI) Oct '24 – May '25
- Student Seed Grant award for a Phase-2 longitudinal project on application-based research funding lab to support undergraduate-led project
 - Developing cyclic peptide therapeutic for Multiple Sclerosis
- Travel Award – National Contraception Meeting May '25
- Awarded by the Male Contraceptive Initiative for travel to Durham, NC, supporting a conference presentation
- NJIT Honors Summer Research Institute (HSRI) Jun '24 – Jul '24
- Grant award by ADHC and URI providing funds to support undergraduate-led projects
 - A combination of Multiple Sclerosis project and Male Contraceptive project
 - Personal compensation included

PUBLICATIONS

Peer-Reviewed

- C. Heffernan, O. West, **K. Singh**, J. Dodd-o, M. James, A. Baradi, K. Rybak, A. Lateef, S. Elkabes, V. Kumar. A novel cyclic peptide that blocks human autoimmune antibodies for the treatment of multiple sclerosis. *European Journal of Medicinal Chemistry*, 119173 (2026). doi:10.1016/j.ejmech.2026.119173

Manuscripts in Preparation

- **K. Singh**, C. Heffernan, O. West, J. Dodd-o, V. Kumar. Machine learning for rational design of a ***** targeting cyclic peptide-based male contraceptive. Publication [in preparation], **Patent application** [in preparation], 2026
- J. Lau, D. Yun, **K. Singh**, A. Patananan, N. Jadavji. Academic Publishing as a Launchpad: How Publishing Research in an Undergraduate Journal Shapes Future Careers. Publication [in preparation], 2026

PRESENTATIONS

M. Kotta, G. Victorio, G. Karmacharya, **K. Singh**, N. Giovannetti, B. Shadpoor, J. Dodd-O, V. Kumar. Putative computational design of novel inhibitors to modulate de-novo targets for CB1 receptor.

UH (Undergraduate Research Day)

Poster + Podium 2026

K. Singh, C. Heffernan, J. Dodd-O, V. Kumar. Development of a novel male contraceptive using cyclic peptides to inhibit fertilization.

AAP/ASCI/APSA Joint Meeting

Poster 2026

NJIT AI Exploration Day

Poster 2026

NJIT ADHC (Honors Interdisciplinary Research Forum) **First Place**

Poster Fall 2025

NCM (National Contraception Meeting)

Poster 2025

NJIT ADHC (Honors Interdisciplinary Research Forum)

Poster Spring 2025

NJIT (Dana Knox Research Showcase)

Poster 2025

NJIT (Undergraduate Research & Innovation)

Abstract + Podium 2025

BMES (Biomedical Engineering Society) [Co-Presenter]

Abstract + Poster 2024

NJIT (Undergraduate Research & Innovation)

Abstract + Podium 2024

K. Singh, J. Lau, H. Gerber, A. Alibilli, L. Goralsky. Mitigating Knowledge Drain: A Case Study in Optimizing Workflow and Training in a High-Turnover, International Peer-Reviewed Undergraduate Science Academic Journal.

NCUR

Poster 2026

ACADEMIC SERVICE

22nd Editor-in-Chief, Associate Editor (Former), Journal of Young Investigators

Dec '24 – Present

- Manage timely publication of high-quality scientific content
- Develop initiatives to further the journal's mission
- Former specialization in Biology, Physical Sciences, and Social Science Departments

Research Lead, Governor's STEM Scholars, Research & Development Council of New Jersey

Sep '26 – May '27

- Guiding a team of high school Scholars through a year-long research and engineering design project anchored to a UN Sustainable Development Goal and addressing a New Jersey community need
- Overseeing team meetings, deliverables, timeline, and budget, and preparing the team's final presentation across four statewide conferences on STEM in government, academia, and industry
- Appointment within a public-private partnership of the Office of the Governor, NJ Department of Education, and NJ Office of the Secretary of Higher Education; \$1,000 completion stipend

Research Chair, American Association of Physicians of Indian Origin (AAPI) PreMed Board

Jun '25 – Present

- Lead committee and organize events to achieve research-related objectives on a national level to support students, researchers, physicians, and more of an Indian origin

Board Member, MCI Youth Advisory Board (YAB)

Aug '25 – Present

Member, Pre-Medical Ad Hoc Committee, APSA

Mar '26 – Present

Member, Goldwater Ambassadors Program

Aug '25 – Present

- Selected into competitive process as a mentee, mentored by Goldwater Scholars
- Program specially designed for students from underrepresented or underserved backgrounds

Member, ARTISAN Program

Apr '25 – Present

- Inducted in the NSF ART Integrated Student Ambassador Network (ARTISAN) program, based off commitment to translational research in my work

Member, Inclusive Excellence Student Advisory Board

Feb '25 – Present

Member, Multifaith Council

Jan '25 – Present

Member, Student Conduct Board

Dec '24 – Present

- Serve as adjudicator on academic and behavioral cases in student conduct hearings, along with other members of the board, including faculty and NJIT administration

Chair, Chair-in-Training (Former), Highlander Integrity Council	Sep '24 – Present
Organizer & Panelist, 'It's Different for Girls' Documentary Screening & Panel, NJIT ADHC & CSLA Co-Sponsored Colloquium	Mar '26
NJIT Representative, BioNJ 33 rd Annual Dinner Meeting & Innovation Celebration	Feb '26
Grant Reviewer, MCI YAB Undergraduate Project Grant	Dec '25 – Jan '26
Teaching Assistant, Honors First Year Seminar	Sep '24 – Dec '24, Sep '25 – Dec '25
Invited Speaker, NJIT First Aid Squad	Sep '25, Oct '25
– Presented to NJIT Board of Trustees as well as the President's Town Hall with squad leadership at the request of NJIT President Teik Lim	
Steering Committee Lead, ADHC First Year Retreat	Mar '25 – Aug '25
Teaching Assistant, Grace Hopper AI Research Institute	May '25 – Aug '25
– Computational Protein Design	
Featured Speaker, ADHC Advising Day	Jun '24 & '25
Advisor, President (Former), Secretary (Former), Honors Hall Council	Feb '24 – May '25
ADHC Representative, National Collegiate Honors Council Annual Conference	Nov '24

LECTURES & FEATURES

- Lecture Series. Ethics in Science and Research, Science Mentorship Institute
- Session Moderator. Student Oral Presentation (Communications). NCUR, 2026
- Featured Presenter. 'Navigating peer review and the student journal landscape'. "Shaping Asia's Future: Development, Innovation, and Society in Transition" organized by Nazarbayev University, Asian Universities Alliance 2026
- Featured Panelist. 'It's Different for Girls' Documentary Screening & Panel. NJIT ADHC & CSLA Co-Sponsored Colloquium, NJIT 2026
- NJIT Makes: Undergraduate Research & Innovation, NJIT Promotional Material. NJIT, 2026
- Albert Dorman Honors College, Honors in Action 2025 (Research, Award, and Study Abroad Experience). ADHC Publication. NJIT, 2025
- Nayib Morán. '150+ Honors Scholars Showcase Findings at Interdisciplinary Research Forum'. Article Feature. NJIT News, 2025
- Albert Dorman Honors College, Honors in Action 2024 (Research and Study Abroad Experience). ADHC Publication. NJIT, 2024
- Gurnoor Kaur. 'Undergraduate Student Researchers on Campus'. Article Feature. NJIT (Student Newspaper – The Vector), 2024
- K. Singh. 'Introduction to PyMOL and Linear Peptide Design'. Lecture. NJIT (Genetically Engineered Machines), 2024

CERTIFICATIONS & LICENSES

- New Jersey EMT License
- ACS "Stop the Bleed" Certification & Instructor License
- OSHA General Industry Safety & Health Certification
- CITI Biomedical Research Certification
- CITI RCR Basic Certification
- American Heart Association BLS & CPR for Adult/Child/Infant Certification

LEADERSHIP

Ethics Coordinator, Mentor (Neuro + Bio), Ethics Curriculum Dev (Former), Admission Officer, Symposium Committee Organizer, Grant Writer, Science Mentorship Institute (Sci-Mi)	May '24 – Present
Public Relations Chair, EMT, NJIT First Aid Squad	Oct '25 – Present, Jan '24 – Present
Stop the Bleed Instructor (Hybrid English–Punjabi), NJ Gurdwaras, NJ Office of Homeland Security & Preparedness (NJOHSP) & Sikh Youth Alliance	Jan '26 – Present
Khalsa School Virsa (Sikh History) Teacher, Sikh Sabha of New Jersey	Sep '20 – Present
Youth Camp Organizer & Coordinator, Sikh Sabha of New Jersey	Aug '21 – Present

Content Manager, The Mardana Project	May '26 – Present
Youth Camp Counselor, Guru Nanak Sikh Heritage	Aug '21 – Aug '25
President, Events Coordinator (Former), NJIT Sikhs	Dec '23 – May '26
President, Treasurer (Former), Co-Founder, NJIT MedicZero	Nov '23 – May '26
Co-Founder, Delta Epsilon Mu (Pre-medical Professional Fraternity)	Feb '24 – May '25
Fellow, Honors Summer Research Institute Fellowship	Jun '24 – Jul '24, Jun '25 – Jul '25
EMT Senior Squad Member, Cadet Crew Chief (Former), Marlboro First Aid Squad	Apr '22 – Jan '25
Fellow, No Patient Left Behind Fellowship	Jun '24 – Aug '24
Young Scholar, American College of Cardiology	May '23 – Apr '24
<i>References and letters of recommendation available upon request.</i>	